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Comptia A+ Assignment

Module -1, 2 [Hardware and its components]

1.What is input device?

> an input device is a hardware device used to enter data

Example

Keyboard

Mouse

Scanner webcam.

2.What are output device?

> an output device shows the result from the computer

Example

Moniter

Printer

Speaker

3.What is CPU?

> central process unit is brain of a computer.

4.What are the types of CPU?

> single core

Duo core
Quad core
Hexa core
Octa core

5. What do we need to keep the CPU Healthy?

- > clean dust regular
 - Use thermal paste
 - Keep cooling fan
 - Avoid over heating
 - Use power supply

6. Do a practical to remove processor and apply thermal paste in it and install it again

- > shut down computer and unplug power.
- Open cabinet.
- Remove cpu cooling fan.
- Unlock processor socket and remove cpu carefully.
- Clean old thermal.
- Apply small amount thermal paste.
- Reinstall cpu.
- Lock socket and fan.
- Start system.

7. Do a practical to Identify CPU and its Sockets.

- > open cabinet.

Search processor on motherboard.

Check socket type written near cpu.

Example of socket is LGA1151 AND AM4.

8.What is memory?

> memory store data and instruction for the computer.

9.What are the types of memory?

> primary memory

RAM and ROM

Secondary memory

Hard disk

Ssd

Pen drive.

10.Do a practical to identify memory types.

> open cabinet

Locate ram slot

Check ram label

Identify storage device

11.Do a practical to install memories in system

>turn off computer

Open ram clips

Alind ram

Press ram gently

Start computer and check

12. Do a practical to identify main memory frequencies.

> open bios

Check ram frequency

Eg. 2400 or 3200 mhz.

13. What is bios

> basic input output system is a firmware stored on motherboard

It starts computer and checks hardware.

14. Describe working process of BIOS.

> 1. Power ON computer.

2. BIOS performs POST (Power On Self Test).

3. Detects hardware devices.

4. Loads bootloader from storage device.

5. Starts operating system.

15. Do a practical to reset bios when system is on.

> 1. Restart computer.

2. Enter BIOS setup using Del/F2 key.

3. Select "Load Default Settings."

4. Save and exit.

16. Do a practical of Hard resetting the BIOS.

> 1. Turn off computer and unplug power.

2. Remove CMOS battery for 5 minutes.
3. Reinsert battery.
4. Start computer.

17. Do a practical of identifying BIOS chip from the motherboard □ What is CMOS?

- > 1. Open motherboard cabinet.
- 2. Locate small chip labeled BIOS/UEFI.
- 3. Usually near CMOS battery.

And CMOS stores BIOS settings like date, time, and boot settings.

18. What is motherboard?

> Motherboard is the main circuit board connecting all computer parts.

19. Describe types of motherboard.

> ATX

Micro-ATX

Mini-ITX

20. Do a practical by identifying parts of motherboard.

> Identify:

CPU socket

RAM slots

PCI slots

SATA ports

CMOS battery

BIOS chip

21. Do a practical by removing all removable parts from the motherboard.

> . Power off system.

2. Remove RAM.

3. Remove CPU fan.

4. Remove storage cables.

5. Remove expansion cards carefully.

22. What is system bus

> System bus transfers data between CPU, memory, and devices.

23. What is chipset and types of chipset?

> Chipset controls communication between components.

Types:

Northbridge

Southbridge

24. Describe how does the Northbridge chipset work what is SMPS? And its purpose DO a practical to install SMPS.

> Northbridge connects CPU with:

* RAM

* Graphics card

* High-speed devices

SMPS (Switched Mode Power Supply) converts AC power into DC power for computer parts.

Practical

1. Open cabinet.
2. Place SMPS in cabinet slot.
3. Tighten screws.
4. Connect motherboard and device power cables.

25. How to check smps?

- > * Check fan rotation
- * Use SMPS tester or multimeter
- * Check if system powers ON

26. List out the types of storage devices.

- > HDD
- * SSD
- * Pen Drive
- * CD/DVD
- * Memory Card

27. Describe the working process of storage devices.

- > Storage devices save data permanently and provide data when needed.

28. Do a practical to Remove storage devices and reinstall it and make a gpt disk.

- >1. Remove SATA/power cable.
- 2. Remove storage drive.
- 3. Reinstall drive and connect cables.
- 4. Open Disk Management.
- 5. Convert disk to GPT format.

29.What is SATA?

> SATA (Serial ATA) is an interface used to connect storage devices

30.Describe the working of SATA.

>SATA transfers data between motherboard and storage devices.

31.Do a practical to install SATA.

- > 1. Connect SATA data cable to motherboard.
- 2. Connect SATA power cable from SMPS.
- 3. Start computer and check detection.

32.What is SCSI storage and type of scsi

> CSI (Small Computer System Interface) is a storage interface used in servers.

Types:

- * Parallel SCSI
- * Serial Attached SCSI (SAS)

33.What is I/O ports?

>I/O ports connect external devices to computer.

34.List out the I/O ports available Do a practical to identify the I/O ports □
What is Boot Process?

>* USB

* HDMI

* VGA

* Ethernet

* Audio port

* PS/2 port

Practical:

Identify ports on motherboard or cabinet.

Boot process is the process of starting the computer and loading the operating system.

35.Describe the boot process in Linux? □ List out the types of display?

>1. BIOS/UEFI starts system

2. GRUB bootloader loads

3. Linux kernel loads

4. System services start

5. Login screen appears

* CRT

* LCD

- * LED
- * OLED

36. What is printer? And type of printer

> Printer is an output device used to print documents.

Types:

- * Inkjet Printer
- * Laser Printer
- * Dot Matrix Printer

37. Do a practical to install the printer

- > 1. Connect printer to computer.
- 2. Install printer driver.
- 3. Add printer in settings.
- 4. Print test page.

38. Do a practical to Troubleshoot the improper printing.

- > 1. Check paper and ink/toner.
- 2. Restart printer.
- 3. Check cable connection.
- 4. Reinstall printer driver.
- 5. Clean print head if needed.

39. What are the parts of laptop?

- > * Display

- * Keyboard
- * Touchpad
- * Battery
- * Motherboard
- * RAM
- * Storage
- * Cooling fan

40. Do a practical to disassemble the laptop.

>

1. Turn off laptop and remove charger.
2. Remove battery if removable.
3. Open screws carefully.
4. Remove back cover.
5. Remove RAM, storage, fan, etc. carefully.
6. Reassemble properly.



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